Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/20/2005
PI Name:	Helble, Joseph J. Ph.D.		
Project Title:	High Capacity Thermally Regenerated Supported Amines for CO2 Capture		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Joseph.J.Helble@Dartmouth.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	603-646-2238
Organization Name:	Darthmouth University		
PI Address 1:	Thayer School of Engineering		
PI Address 2:	8000 Cummings Hall		
PI Web Page:			
City:	Hanover	**State:**	NH
Zip Code:	03755-8000	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	08/02/2002	**End Date:**	09/01/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Zeolite sorbents are presently used for CO2 removal on long-term human-occupied space missions. These sorbents are regenerable, but the regeneration process is energy intensive. The focus of this project is therefore to investigate novel supported amine sorbents that can be regenerated at much lower temperatures and with lower energy than presently required by zeolites. These sorbents will be prepared by immobilizing amines on polymeric supports, chemically modifying the amines to maximize CO2 capture capacity, and demonstrating the efficacy of a temperature swing process as the regeneration step. The temperature swing process has the potential to provide a major reduction in the energy required for sorbent regeneration, an important benefit in space environments. Supported amines also have the potential to reduce the energy costs associated with the pumped liquid systems envisioned for terrestrial flue gas CO2 removal If successful, this project is ultimately expected to lead to the development of commercial systems for high capacity reduced cost CO2 removal, suitable for both enclosed habitats and terrestrial applications. Develop low cost		

	CO2 sorbents that require reduced energy consumption for thermal regeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The focus of this project is to investigate novel supported amine sorbents that can be regenerated at much lower temperatures and with lower energy than presently required by zeolites. The proposed use of a temperature swing regeneration process has the potential to provide a major reduction in the energy required for sorbent regeneration, an important benefit in space environments. Supported amines also have the potential to reduce the energy costs associated with the pumped liquid systems envisioned for terrestrial flue gas CO2 removal If successful, this project is thus expected to lead to the development of commercial systems for high capacity reduced cost CO2 removal, suitable for both enclosed habitats and terrestrial applications.
Task Progress:	Taguchi statistical design of experiment methods were used to identify the main variables influencing the thermal swing (TSA) process. Included in the experiment were the following seven variables: cycle time (15-20 min), desorption pressure (0.5-1.7 psia), desorption temperature (50-60°C), inlet concentration of CO2 (5-10 mmHg), residence time (1.7-3 sec), absorption temperature (20-30°C), and dew point (0-24°C). There were eight conditions that were specified by the Taguchi method for identification of which of the seven variables were most important to the system. Each of the eight conditions was examined twice. Previous results reported from experiments conducted with the supported amine tetraethylenepentamine (TEPAN) showed that for the ranges of variables tested, cycle time, dew point, and residence time were not significant to the variation of the system. Desorption pressure showed the greatest variation (61% of the total variation) while desorption and adsorption temperatures showed 19% and 6%, respectively. Inlet CO2 concentration showed a 12% variation of the system. The newest amine, E-100, is a longer chained amine and therefore has more active amine sites and less volatility than TEPAN. Taguchi results from E-100 show less variation for the seven variables tested. Only desorption pressure statistically altered the variation in the system. Future research is focused on optimizing the molecular structure of the amine to enhance cyclic CO2 capacity. Branched and cyclic molecular structures are being considered. The amines that are found to have potential will be tested under the Taguchi method to find the significant variables
Bibliography Type:	Description: (Last Updated: 08/11/2011)
Papers from Meeting Proceedings	Schladt M, Filburn,T, Helble JJ. "High Capacity Thermally Regenerated Supported Amine Sorbents for CO2 Removal." Presented at the 34th International Conference on Environmental Systems (ICES), Colorado Springs, CO, July 2004. 34th International Conference on Environmental Systems (ICES), Colorado Springs, CO, July 2004. SAE Paper 2004-01-2443. http://dx.doi.org/10.4271/2004-01-2443 , Jul-2004